

・分数関数の不定積分～部分分数分解～

(例1) 次の不定積分を求めよ。

$$(1) \int \frac{1}{x^2-1} dx \quad (2) \int \frac{x+4}{x^2-x-2} dx$$

point

部分分数分解して $\frac{\text{定数}}{\text{1次式}} \cdot \frac{f'(x)}{f(x)}$ をつくる。

Cは積分定数とする。

$$\begin{aligned} (1) \int \frac{1}{x^2-1} dx &= \frac{1}{2} \int \left(\frac{1}{x-1} - \frac{1}{x+1} \right) dx \\ &= \frac{1}{2} (\log|x-1| - \log|x+1|) + C \\ &= \frac{1}{2} \log \left| \frac{x-1}{x+1} \right| + C \quad // \end{aligned}$$

$$\begin{aligned} (2) \int \frac{x+4}{x^2-x-2} dx &= \int \left(-\frac{1}{x+1} + \frac{2}{x-2} \right) dx \\ &= -\log|x+1| + 2\log|x-2| + C \\ &= \log \frac{(x-2)^2}{|x+1|} + C \quad // \end{aligned}$$

$$\begin{aligned} \frac{x+4}{(x+1)(x-2)} &= \frac{a}{x+1} + \frac{b}{x-2} \\ a(x-2) + b(x+1) &= x+4 \\ \begin{cases} -2a = 1 \\ 3b = 6 \end{cases} &\quad \therefore \begin{cases} a = -1 \\ b = 2 \end{cases} \end{aligned}$$

(例2) 次の不定積分を求めよ。

Cは積分定数とする。

$$\begin{aligned} (1) \int \frac{x^2-2x-2}{x^3-1} dx &= \int \left(-\frac{1}{x-1} + \frac{2x+1}{x^2+x+1} \right) dx \quad \leftarrow \begin{aligned} &\frac{x^2-2x-2}{x^3-1} = \frac{a}{x-1} + \frac{bx+c}{x^2+x+1} \\ &a(x^2+x+1) + (bx+c)(x-1) = x^2-2x-2 \\ &\begin{cases} 3a = -3 \\ a-c = -2 \\ a-2(-b+c) = 1 \end{cases} \quad \therefore \begin{cases} a = -1 \\ b = 2 \\ c = 1 \end{cases} \end{aligned} \\ &= \int \left(-\frac{1}{x-1} + \frac{(x^2+x+1)'}{x^2+x+1} \right) dx \\ &= -\log|x-1| + \log|x^2+x+1| + C \quad \leftarrow x^2+x+1 = \left(x+\frac{1}{2}\right)^2 + \frac{3}{4} > 0 \\ &= \log \left| \frac{x^2+x+1}{x-1} \right| + C \quad // \end{aligned}$$

$$\begin{aligned} (2) \int \frac{1}{x(x+1)(x+2)} dx &= \int \left(\frac{1}{2} \cdot \frac{1}{x} - \frac{1}{x+1} + \frac{1}{2} \cdot \frac{1}{x+2} \right) dx \quad \leftarrow \begin{aligned} &\frac{1}{x(x+1)(x+2)} = \frac{a}{x} + \frac{b}{x+1} + \frac{c}{x+2} \\ &a(x+1)(x+2) + bx(x+2) + cx(x+1) = 1 \\ &\begin{cases} 2a = 1 \\ -b = 1 \\ 2c = 1 \end{cases} \quad \therefore \begin{cases} a = \frac{1}{2} \\ b = -1 \\ c = \frac{1}{2} \end{cases} \end{aligned} \\ &= \frac{1}{2} (\log|x| - 2\log|x+1| + \log|x+2|) + C \\ &= \frac{1}{2} \log \left| \frac{x(x+2)}{(x+1)^2} \right| + C \quad // \end{aligned}$$

$$\begin{aligned} (3) \int \frac{x^2}{(x+1)^2(x-1)} dx &= \int \left(\frac{3}{4} \cdot \frac{1}{x+1} - \frac{1}{2} \cdot \frac{1}{(x+1)^2} + \frac{1}{4} \cdot \frac{1}{x-1} \right) dx \quad \leftarrow \begin{aligned} &\frac{x^2}{(x+1)^2(x-1)} = \frac{a}{x+1} + \frac{b}{(x+1)^2} + \frac{c}{x-1} = x^2 \\ &a(x+1)(x-1) + b(x-1) + c(x+1)^2 = x^2 \\ &\begin{cases} -2b = 1 \\ 4c = 1 \\ -a-b+c = 0 \end{cases} \quad \therefore \begin{cases} a = \frac{3}{4} \\ b = -\frac{1}{2} \\ c = \frac{1}{4} \end{cases} \end{aligned} \\ &= \frac{3}{4} \log|x+1| - \frac{1}{2} \left(-\frac{1}{x+1} \right) + \frac{1}{4} \log|x-1| + C \\ &= \frac{1}{4} \log|(x+1)^3(x-1)| + \frac{1}{2(x+1)} + C \quad // \end{aligned}$$